

PROJECT TITLE :

“Toy Shop Management System: A Seller and Buyer Program”

PROBLEM DESCRIPTION / OVERVIEW :

This program is a command-line-based toy store system where users can act as either a Seller or a Buyer.

- The Seller can add toys, set prices, and manage stock.
- The Buyer can view available toys and purchase them using a balance.
- The system uses a dictionary (I) to store product data such as price and stock.
- The program runs in a loop until the user chooses to exit.

It simulates a simple inventory management + shopping system.

HOW TO RUN THE PROGRAM :

1. Make sure you have Python installed (Python 3.x)
2. Copy the code into a .py file. example: toy_store.py
3. Run the program using: python toy_store.py
4. Follow the on-screen menu:
Choose Seller (a) → manage inventory
Choose Buyer (b) → buy items
Choose Exit (c) → stop program

SAMPLE INTERACTION / EXAMPLE OUTPUT :

Start:

Pick a Role:

a. Seller

b. Buyer

c. Exit

Input: a

Seller adds product:

----[TOY MAKER]----

Enter a toy name: Teddy

Enter price: 100

Enter quantity: 5

[----Added Teddy to inventory]----

Buyer view:

----[SHOP]----

Balance: 1000

----[AVAILABLE PRODUCTS]----

Toy Name: Teddy

Price: 100

Stock: 5

Buyer purchase:

Enter the name of the toy to buy: Teddy

Successfully bought Teddy for \$100

Remaining Balance: \$900

If not enough balance:

Not enough balance to buy this toy

If out of stock:

Toy is out of stock

REQUIREMENTS

- **Python 3.x**
- **Works in: Terminal / Command Prompt, VS Code / IDLE / PyCharm**

PROBLEM STATEMENT :

Managing a small toy shop manually can make it difficult to keep track of available products, prices, stock quantities, and the buyer's remaining balance. Without a simple system, sellers may have difficulty updating inventory, while buyers may not know which products are available or whether they have enough money to purchase them.

This project aims to create a simple Python-based toy shop system where users can act as either a **Seller** or a **Buyer**. The system will allow sellers to add and update products while allowing buyers to view and purchase available toys using a given balance.

PROBLEM OBJECTIVES :

- General Objective** - To develop a simple and interactive Toy Shop Management System using Python that manages toy inventory and allows users to buy and sell products.
- Specific Objectives**
 - To allow users to choose between Seller and Buyer roles.

- To allow sellers to add new toys to the inventory.
- To allow sellers to update the price and stock of existing toys.
- To display the total number of toys currently in stock.
- To allow buyers to view available toys and their prices.
- To check if the buyer has enough balance to purchase a toy.
- To automatically decrease the toy's stock after a successful purchase.
- To automatically deduct the purchase price from the buyer's balance.
- To provide error messages for invalid inputs and unavailable products

PROBLEM FEATURES :

1. **Role Selection** - Users can choose to become a Seller, Buyer, or Exit the program.
2. **Toy Inventory** - Stores toy names, prices, and quantities using a dictionary.
3. **Add Product** - Sellers can add a new toy to the inventory.
4. **Update Product** - Sellers can update an existing toy's price and stock.
5. **Stock Counter** - Displays the total number of toys currently available.
6. **Product Display** - Buyers can see the available toys, prices, and stock
7. **Balance System** - The buyer starts with a set amount of money.
8. **Purchase System** - Buyers can purchase toys if they have enough money and the product is in stock.
9. **Stock Deduction** - Stock decreases by one after a successful purchase.
10. **Balance Deduction** - The toy's price is deducted from the buyer's balance.
11. **Input Validation** - Prevents invalid numerical inputs and invalid menu choices.
12. **Exit System** - Allows the user to safely exit the program.

PLANNED INPUTS AND OUTPUTS :

A. **INPUTS** - The program will receive the following inputs from the user :

I. Role Selection

A → Seller

B → Buyer

C → Exit

II. Seller inputs

- Toy Name
- Toy Price
- Toy Quantity
- Menu Choices

III. Buyer inputs

- Selected toy name
- Menu Choices

B. **OUTPUTS** - The Program will display :

- Seller and Buyer menus=
- Current inventory
- Total stock
- Toy names
- Toy prices
- Toy quantities
- Buyer's current balance
- Successful purchase messages
- Remaining balance after purchasing
- Error messages for invalid inputs
- Messages for unavailable or out-of-stock toys
- Exit message

FLOWCHART :

